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Centrifugal compressor

Abstract

A centrifugal compressor includes a rotary shaft, a compression mechanism that compresses fluid, a motor, an inverter, a housing, and a wiring unit. The compression mechanism includes an impeller, a suction chamber, a diffuser, and a volute. The housing has a motor housing member, an impeller housing member, a suction housing member, and an inverter housing member. The housing has a through hole that extends to an inside of the inverter housing member from the motor housing member. The wiring unit is disposed in the housing and inserted in the through hole. The through hole is disposed away from the volute. A suction passage is located away from the through hole in a circumferential direction of the rotary shaft.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2024-092251 filed on Jun. 6, 2024, the entire disclosure of which is incorporated herein by reference.

BACKGROUND ART

- (2) The present disclosure relates to a centrifugal compressor.
- (3) The centrifugal compressor includes a rotary shaft, a compression mechanism, a motor, an inverter, and a housing. The compression mechanism compresses fluid with rotation of the rotary shaft. The motor rotates the rotary shaft. The inverter drives the motor. The motor, the compression mechanism, and the inverter are arranged in this order in an axial direction of the rotary shaft and

accommodated in the housing. The compression mechanism has an impeller, a suction chamber, a diffuser, and a volute. The impeller rotates integrally with the rotary shaft. The diffuser is disposed downstream of the impeller in a direction in which the fluid flows. The volute communicates with an outlet of the diffuser and is located outward with respect to the diffuser in a radial direction of the rotary shaft.

(4) The suction chamber may be located closer to the inverter than the impeller 25 in the axial direction of the rotary shaft. As such, when the suction chamber may be located closer to the inverter than the impeller in the axial direction of the rotary shaft, heat generated from the inverter is dissipated to the fluid in the suction chamber. As a result, the inverter is efficiently cooled by the fluid in the suction chamber. In addition, the centrifugal compressor includes a wiring unit electrically connecting the motor to the inverter. For example, in Japanese Patent Application Publication No. H03-111700, a part of the wiring unit extends outside the housing.

(5) As seen in the Publication, when the wiring unit extends outside the housing, a sealing member may be required between the wiring unit and the housing, and the wiring unit may need to be fixed to the housing such that the wiring unit does not interfere with auxiliary equipment, or the like disposed around the centrifugal compressor. These may increase the number of components of the centrifugal compressor and reduce ease of installation of the centrifugal compressor. Accordingly, it is desired to electrically connect the motor to the inverter without increasing the number of components of the centrifugal compressor and reducing the ease of installation of the centrifugal compressor while the centrifugal compressor is prevented from increasing in size.

SUMMARY

(6) In accordance with an aspect of the present disclosure, there is provided a centrifugal compressor that includes a rotary shaft, a compression mechanism that compresses fluid with rotation of the rotary shaft, a motor that rotates the rotary shaft, an inverter that drives the motor, a housing in which the motor, the compression mechanism, and the inverter arranged in this order in an axial direction of the rotary shaft are accommodated, and a wiring unit through which the motor and the inverter are electrically connected to each other. The compression mechanism includes an impeller that rotates integrally with the rotary shaft, a suction chamber that is located closer to the inverter than the impeller in the axial direction, a diffuser that is disposed downstream of the impeller in a direction in which the fluid flows, and a volute that communicates with an outlet of the diffuser and is located outward with respect to the diffuser in a radial direction of the rotary shaft. The housing has a motor housing member in which the motor is accommodated, an impeller housing member in which the impeller is accommodated and that define the diffuser and the volute, a suction housing member that defines the suction chamber, and an inverter housing member in which the inverter is accommodated. The motor housing member, the impeller housing member, the suction housing member, and the inverter housing member are arranged in this order in the axial direction. The housing has a through hole that extends in the axial direction to an inside of the inverter housing member from the motor housing member through the impeller housing member and the suction housing member. The wiring unit is disposed in the housing and inserted in the through hole. The through hole is disposed away from the volute. A suction passage through which the suction chamber is connected to an outside of the housing is located away from the through hole in a circumferential direction of the rotary shaft.

(7) Other aspects and advantages of the disclosure will become apparent from the following description, taken in conjunction with the accompanying drawings, illustrating by way of example the principles of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure, together with objects and advantages thereof, may best be understood by reference to the following description of the embodiments together with the accompanying drawings in which:

(2) FIG. 1 is a cross-sectional view illustrating a centrifugal compressor according to an embodiment;

(3) FIG. 2 is an enlarged cross-sectional view of a part of the centrifugal compressor;

(4) FIG. 3 is a cross-sectional view taken along a line III-III of FIG. 2; and

(5) FIG. 4 is a cross-sectional view taken along a line IV-IV of FIG. 2.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(6) The following will describe an embodiment of a centrifugal compressor according to the present disclosure with reference to FIGS. 1 to 4. The centrifugal compressor of the present embodiment is mounted on a vehicle, which is not illustrated, and used in a refrigerating cycle of a vehicle air conditioner. The centrifugal compressor compresses a refrigerant as fluid.

(7) <Basic Configuration of Centrifugal Compressor>

(8) As illustrated in FIG. 1, a centrifugal compressor 10 includes a rotary shaft 11, a compression mechanism 12, a motor 13, an inverter 14, and a housing 15. The compression mechanism 12 has a first impeller 16 as the impeller, a suction chamber 17, a first diffuser 18 as the diffuser, and a first volute 19 as the volute. The compression mechanism 12 compresses a refrigerant with rotation of the rotary shaft 11. The motor 13 rotates the rotary shaft 11. The inverter 14 drives the motor 13.

(9) The housing 15 is formed in a tubular shape. The housing 15 has a motor housing member 21, an impeller housing member 22, a suction housing member 23, and an inverter housing member 24. The housing 15 also has a first plate 25, a second plate 26, a third plate 27, and a fourth plate 28. The motor housing member 21, the impeller housing member 22, the suction housing member 23, the inverter housing member 24, the first plate 25, the second plate 26, the third plate 27, and the fourth plate 28 are made of a metal material, for example, aluminum.

(10) The motor housing member 21 has an end wall 21a formed in a plate shape, a first peripheral wall 21b formed in a tubular shape, and a second peripheral wall 21c formed in a tubular shape. The end wall 21a has a shaft insertion hole 21d formed in a circular hole shape. The shaft insertion hole 21d is formed in a center portion of the end wall 21a. The shaft insertion hole 21d extends through the end wall 21a in a thickness direction of the end wall 21a. The first peripheral wall 21b extends from an outer peripheral portion of one surface of the end wall 21a in the thickness direction of the end wall 21a. The second peripheral wall 21c extends from an outer peripheral portion of the other surface of the end wall 21a in the thickness direction of the end wall 21a.

(11) As illustrated in FIG. 2, the impeller housing member 22 has a first housing member 29 and a second housing member 30. The first housing member 29 is formed in a plate shape. The first housing member 29 is connected to the motor housing member 21 while closing an opening of the first peripheral wall 21b. The end wall 21a and the first peripheral wall 21b of the motor housing member 21, and the first housing member 29 of the impeller housing member 22 define a motor chamber S1. The motor 13 is accommodated in the motor chamber S1. That is, the motor 13 is accommodated in the motor housing member 21.

(12) The first housing member 29 has a shaft insertion hole 29a formed in a circular hole shape. The shaft insertion hole 29a is formed in a center portion of the first housing member 29. The shaft insertion hole 29a extends through the first housing member 29 in a thickness direction of the first housing member 29. An axial line of the shaft insertion hole 29a is aligned with an axial line of the shaft insertion hole 21d.

(13) The second housing member 30 is formed in a plate shape. The second housing member 30 is connected to an end surface of the first housing member 29 with the first housing member 29 interposed between the second housing member 30 and the motor housing member 21. The second

housing member **30** is connected to the first housing member **29** with a thickness direction of the second housing member **30** coinciding with the thickness direction of the first housing member **29**.

(14) The second housing member **30** has a first suction port **31** formed in a circular hole shape. The first suction port **31** is formed in a center portion of the second housing member **30**. The first suction port **31** is opened in one end surface of the second housing member **30**, which is opposite to the other end surface of the second housing member **30** facing the first housing member **29**. The second housing member **30** is connected to the first housing member **29** with an axial line of the first suction port **31** aligned with the axial line of the shaft insertion hole **29a**.

(15) A first impeller chamber **32** is formed inside the first housing member **29** and the second housing member **30**. Accordingly, the first housing member **29** and the second housing member **30** define the first impeller chamber **32**. The first impeller chamber **32** communicates with the first suction port **31**. The first impeller **16** is accommodated in the first impeller chamber **32**. That is, the first impeller **16** is accommodated in the impeller housing member **22**.

(16) The suction housing member **23** is connected to the end surface of the second housing member **30** with the second housing member **30** interposed between the first housing member **29** and the suction housing member **23**. The suction housing member **23** has a chamber forming recess **33**. The chamber forming recess **33** is opened in an end surface of the suction housing member **23** near the second housing member **30**. The second housing member **30** closes the opening of the chamber forming recess **33**. The chamber forming recess **33** and the second housing member **30** define the suction chamber **17**. That is, the suction housing member **23** defines the suction chamber **17**. The suction chamber **17** communicates with the first suction port **31**.

(17) As illustrated in FIG. 1, the inverter housing member **24** has a housing main body **24a** and a covering member **24b**. The housing main body **24a** is formed in a tubular shape. The housing main body **24a** is connected to an end surface of the suction housing member **23** with the suction housing member **23** interposed between the second housing member **30** and the housing main body **24a**. An opening of the housing main body **24a** at a first end thereof is closed by the one end surface of the suction housing member **23**, which is opposite to the other end surface of the suction housing member **23** facing the second housing member **30**. The covering member **24b** is connected to the housing main body **24a** while closing an opening of the housing main body **24a** at a second end thereof. The housing main body **24a**, the covering member **24b**, and the suction housing member **23** define an inverter chamber **34**. The inverter **14** is accommodated in the inverter chamber **34**. That is, the inverter **14** is accommodated in the inverter housing member **24**.

(18) The first plate **25** is connected to the motor housing member **21** while closing an opening of the second peripheral wall **21c** of the motor housing member **21**. The end wall **21a** and the second peripheral wall **21c** of the motor housing member **21**, and the first plate **25** define a bearing accommodating chamber **35**. The bearing accommodating chamber **35** communicates with the shaft insertion hole **21d**. A thrust bearing **36** is accommodated in the bearing accommodating chamber **35**.

(19) The first plate **25** has a second suction port **37** formed in a circular hole shape. The second suction port **37** is formed in a center portion of the first plate **25**. The second suction port **37** is opened in an end surface of the first plate **25** near the motor housing member **21**. The first plate **25** is connected to the motor housing member **21** with an axial line of the second suction port **37** aligned with the axial line of the shaft insertion hole **21d**.

(20) The second plate **26** is connected to an end surface of the first plate **25** with the first plate **25** interposed between the motor housing member **21** and the second plate **26**. The second plate **26** is connected to the first plate **25** with a thickness direction of the second plate **26** coinciding with a thickness direction of the first plate **25**.

(21) The second plate **26** has a shaft insertion hole **26a**. The shaft insertion hole **26a** is formed in a center portion of the second plate **26**. The shaft insertion hole **26a** extends through the second plate **26** in the thickness direction of the second plate **26**. An axial line of the shaft insertion hole **26a** is

aligned with the axial line of the second suction port 37.

(22) A second impeller chamber 38 is formed inside the first plate 25 and the second plate 26. Thus, the first plate 25 and the second plate 26 define the second impeller chamber 38. The second impeller chamber 38 communicates with the second suction port 37. Furthermore, the second impeller chamber 38 communicates with the shaft insertion hole 26a.

(23) The third plate 27 is connected to an end surface of the second plate 26 with the second plate 26 interposed between the first plate 25 and the third plate 27. The third plate 27 is connected to the second plate 26 with a thickness direction of the third plate 27 coinciding with the thickness direction of the second plate 26.

(24) The third plate 27 has a third suction port 39 formed in a circular hole shape. The third suction port 39 is formed in a center portion of the third plate 27. The third suction port 39 is opened in one end surface of the third plate 27, which is opposite to the other end surface of the third plate 27 facing the second plate 26. The third plate 27 is connected to the second plate 26 with an axial line of the third suction port 39 aligned with the axial line of the shaft insertion hole 26a.

(25) A third impeller chamber 40 is formed inside the second plate 26 and the third plate 27. Thus, the second plate 26 and the third plate 27 define the third impeller chamber 40. The third impeller chamber 40 communicates with the third suction port 39. Furthermore, the third impeller chamber 40 communicates with the shaft insertion hole 26a.

(26) The fourth plate 28 is connected to an end surface of the third plate 27 with the third plate 27 interposed between the second plate 26 and the fourth plate 28. The fourth plate 28 closes the opening of the third suction port 39 in an axial direction of the third suction port 39.

(27) The rotary shaft 11 is accommodated in the housing 15. A first end of the rotary shaft 11 is located in the first impeller chamber 32. The rotary shaft 11 extends through the shaft insertion hole 29a, the motor chamber S1, the shaft insertion hole 21d, the bearing accommodating chamber 35, the second suction port 37, the second impeller chamber 38, and the shaft insertion hole 26a. A second end of the rotary shaft 11 is located in the third impeller chamber 40.

(28) The first impeller 16 is connected to a first end portion of the rotary shaft 11. The centrifugal compressor 10 includes a second impeller 41 and a third impeller 42. The second impeller 41 is accommodated in the second impeller chamber 38. The third impeller 42 is accommodated in the third impeller chamber 40. The second impeller 41 and the third impeller 42 are connected to a second end portion of the rotary shaft 11. The first impeller 16, the second impeller 41, and the third impeller 42 rotate integrally with the rotary shaft 11.

(29) The first impeller 16 rotates integrally with the rotary shaft 11 in order to compress the refrigerant sucked into the first impeller chamber 32 through the first suction port 31. The second impeller 41 rotates integrally with the rotary shaft 11 in order to compress the refrigerant sucked into the second impeller chamber 38 through the second suction port 37. The third impeller 42 rotates integrally with the rotary shaft 11 in order to compress the refrigerant sucked into the third impeller chamber 40 through the third suction port 39.

(30) The motor housing member 21, the impeller housing member 22, the suction housing member 23, and the inverter housing member 24 are arranged in this order in the axial direction of the rotary shaft 11. The motor 13, the compression mechanism 12, and the inverter 14 are arranged in this order in the axial direction of the rotary shaft 11 and accommodated in the housing 15. The suction chamber 17 is located closer to the inverter 14 than the first impeller 16 in the axial direction of the rotary shaft 11.

(31) A first radial bearing 43 is provided between the shaft insertion hole 21d and the rotary shaft 11. The first radial bearing 43 rotatably supports the rotary shaft 11 in a radial direction thereof. A second radial bearing 44 is provided between the shaft insertion hole 29a and the rotary shaft 11. The second radial bearing 44 rotatably supports the rotary shaft 11 in the radial direction thereof.

(32) The centrifugal compressor 10 includes a thrust collar 45. The thrust collar 45 is formed in a circular plate shape. The thrust collar 45 protrudes from an outer circumferential surface of the

rotary shaft **11**. The thrust collar **45** is press-fitted into the outer circumferential surface of the rotary shaft **11**. The thrust collar **45** rotates integrally with the rotary shaft **11**. The thrust collar **45** is disposed in the bearing accommodating chamber **35**. The thrust bearing **36** rotatably supports the rotary shaft **11** through the thrust collar **45** in a thrust direction of the rotary shaft **11**.

(33) As illustrated in FIG. 2, the first diffuser **18** and the first volute **19** are formed inside the first housing member **29** and the second housing member **30**. That is, the impeller housing member **22** defines the first diffuser **18** and the first volute **19**. The first diffuser **18** is disposed downstream of the first impeller **16** in a direction in which the refrigerant flows. The first volute **19** extends around the axial line of the first suction port **31** so as to surround the first impeller chamber **32**. The first volute **19** communicates with an outlet of the first diffuser **18** and is located outward with respect to the first diffuser **18** in the radial direction of the rotary shaft **11**.

(34) As illustrated in FIG. 1, the centrifugal compressor **10** includes a second diffuser **46** and a second volute **47**. The second diffuser **46** and the second volute **47** are formed inside the first plate **25** and the second plate **26**. Accordingly, the first plate **25** and the second plate **26** define the second diffuser **46** and the second volute **47**. The second diffuser **46** is disposed downstream of the second impeller **41** in the direction in which the refrigerant flows. The second volute **47** extends around the axial line of the second suction port **37** so as to surround the second impeller chamber **38**. The second volute **47** communicates with an outlet of the second diffuser **46** and is located outward with respect to the second diffuser **46** in the radial direction of the rotary shaft **11**.

(35) The centrifugal compressor **10** includes a third diffuser **48** and a third volute **49**. The third diffuser **48** and the third volute **49** are formed inside the second plate **26** and the third plate **27**. Accordingly, the second plate **26** and the third plate **27** define the third diffuser **48** and the third volute **49**. The third diffuser **48** is disposed downstream of the third impeller **42** in the direction in which the refrigerant flows. The third volute **49** extends around the axial line of the third suction port **39** so as to surround the third impeller chamber **40**. The third volute **49** communicates with an outlet of the third diffuser **48** and is located outward with respect to the third diffuser **48** in the radial direction of the rotary shaft **11**.

(36) The centrifugal compressor **10** has a first communication hole **50**. The first communication hole **50** is formed in the first housing member **29**. A first end of the first communication hole **50** communicates with the first volute **19**. A second end of the first communication hole **50** communicates with the motor chamber **S1**. The centrifugal compressor **10** has a plurality of second communication holes **51**. The second communication holes **51** are formed in the end wall **21a** of the motor housing member **21**. A first end of each of the second communication holes **51** communicates with the motor chamber **S1**. A second end of each of the second communication holes **51** communicates with the bearing accommodating chamber **35**.

(37) The centrifugal compressor **10** has a third communication hole **52**. The third communication hole **52** extends through the second plate **26** and the third plate **27**. A first end of the third communication hole **52** communicates with the second volute **47**. The third communication hole **52** is, at a second end thereof, opened in the one end surface of the third plate **27**, which is opposite to the other end surface of the third plate **27** facing the second plate **26**. The centrifugal compressor **10** has a communication groove **53**. The communication groove **53** is formed in the one end surface of the third plate **27**, which is opposite to the other end surface of the third plate **27** facing the second plate **26**. The communication groove **53** extends in the radial direction of the rotary shaft **11**. The communication groove **53** is closed by the fourth plate **28**. The communication groove **53** and the fourth plate **28** define a communication passage **54**. A first end of the communication passage **54** communicates with the second end of the third communication hole **52**. A second end of the communication passage **54** communicates with the third suction port **39**.

(38) The motor **13** includes a stator **55** and a rotor **56**. The stator **55** has a stator core **57** formed in a cylindrical shape and a coil **58** wound around the stator core **57**. The stator core **57** is fixed to an inner peripheral surface of the first peripheral wall **21b** of the motor housing member **21**. The rotor

56 is disposed inside the stator core **57** in the motor chamber **S1**. The rotor **56** rotates integrally with the rotary shaft **11**. The rotor **56** has a rotor core **59** fixed to the rotary shaft **11** and a plurality of permanent magnets provided in the rotor core **59**. Illustrations of the permanent magnets are omitted.

(39) <First Volute>

(40) As illustrated in FIG. 3, the first volute **19** has a scroll passage **60** and a discharge passage **61**. The scroll passage **60** extends spirally around an axial line **L1** of the rotary shaft **11** in a circumferential direction of the rotary shaft **11**. The scroll passage **60** communicates with the outlet of the first diffuser **18**. The scroll passage **60** surrounds the first diffuser **18**. The scroll passage **60** is located outward with respect to the first diffuser **18** in the radial direction of the rotary shaft **11** and extends over an entire length of the rotary shaft **11** in the circumferential direction thereof. The scroll passage **60** has a minimum scroll passage portion **60a** and a maximum scroll passage portion **60b**. The minimum scroll passage portion **60a** is a portion having the minimum cross-sectional area of the scroll passage **60** and the maximum scroll passage portion **60b** is a portion having the maximum cross-sectional area of the scroll passage **60**. The cross-sectional area of the scroll passage **60** gradually increases as the scroll passage **60** extends in the circumferential direction of the rotary shaft **11** from the minimum scroll passage portion **60a** and reaches the cross-sectional area of the maximum scroll passage portion **60b**. The minimum scroll passage portion **60a** and the maximum scroll passage portion **60b** communicate with each other.

(41) The discharge passage **61** branches off from the scroll passage **60**. The discharge passage **61** is located outward with respect to the scroll passage **60** in the radial direction of the rotary shaft **11** and extends around the axial line **L1** of the rotary shaft **11** in the circumferential direction of the rotary shaft **11**. The discharge passage **61** branches off from the maximum scroll passage portion **60b** of the scroll passage **60**. Thus, a connecting point at which the discharge passage **61** is connected to the maximum scroll passage portion **60b** is also a branch point **62** at which the discharge passage **61** branches off from the scroll passage **60**.

(42) Here, when viewed in the axial direction of the rotary shaft **11**, a straight line intersecting the axial line **L1** of the rotary shaft **11** and passing through the branch point **62** at which the discharge passage **61** branches off from the scroll passage **60** is defined as an imaginary straight line **L10**. Furthermore, when viewed in the axial direction of the rotary shaft **11**, a straight line intersecting the axial line **L1** of the rotary shaft **11** and extending perpendicular to the imaginary straight line **L10** is defined as a perpendicular line **L11**. The discharge passage **61** extends toward the perpendicular line **L11** from the branch point **62** located on the imaginary straight line **L10**. The discharge passage **61** extends toward the perpendicular line **L11** from the imaginary straight line **L10** without cutting across the perpendicular line **L11**.

(43) <Suction Chamber>

(44) As illustrated in FIGS. 3 and 4, when viewed in the axial direction of the rotary shaft **11**, the suction chamber **17** is overlapped with the first impeller **16**, the first diffuser **18**, and the first volute **19** such that the suction chamber **17** covers them. As illustrated in FIG. 4, the centrifugal compressor **10** includes a suction passage **63**. The suction passage **63** is formed in the suction housing member **23**. The suction passage **63** extends in the radial direction of the rotary shaft **11**. A first end of the suction passage **63** communicates with an outside of the suction housing member **23**. A second end of the suction passage **63** communicates with the suction chamber **17**. Thus, the suction chamber **17** and an outside of the housing **15** are connected to each other through the suction passage **63**. The refrigerant outside the housing **15** is sucked into the suction chamber **17** through the suction passage **63**.

(45) <Through Hole>

(46) As illustrated in FIG. 2, a through hole **64** is formed in the housing **15**. The through hole **64** extends in the axial direction of the rotary shaft **11** to an inside of the inverter housing member **24** from the motor housing member **21** through the impeller housing member **22** and the suction

housing member **23**.

(47) As illustrated in FIG. 3, the through hole **64** is formed in a rectangular hole shape. The through hole **64** is disposed away from the first volute **19**. The through hole **64** is also disposed away from the suction passage **63**. That is, the suction passage **63** is located away from the through hole **64** in the circumferential direction of the rotary shaft **11**. The through hole **64** is disposed on an extension line of the discharge passage **61** extending in the circumferential direction of the rotary shaft **11**. Note that the extension line of the discharge passage **61** extending in the circumferential direction of the rotary shaft **11** passes through an imaginary circle C1 having a point on the axial line L1 of the rotary shaft **11** as a center. When viewed in the axial direction of the rotary shaft **11**, the through hole **64** is located on the same side as the discharge passage **61** with respect to the imaginary straight line L10 and adjacent to a downstream end of the discharge passage **61** in the circumferential direction of the rotary shaft **11**.

(48) <Wiring Unit>

(49) As illustrated in FIG. 2, the centrifugal compressor **10** includes a wiring unit **65**. The wiring unit **65** has a conductive member **66**, a motor wire **67**, and a cluster block **68**. The conductive member **66** is formed in a columnar shape. The conductive member **66** is supported by the housing **15** through a supporting member **69**. Specifically, the supporting member **69** is formed in a plate shape. The supporting member **69** is disposed in the inverter chamber **34**. The supporting member **69** is fixed to the one end surface of the suction housing member **23**, which is opposite to the other end surface of the suction housing member **23** facing the second housing member **30**. The conductive member **66** is supported by the supporting member **69**. A first end of the conductive member **66** is electrically connected to the inverter **14**. A second end of the conductive member **66** is inserted in the through hole **64**. Thus, the conductive member **66** extends from the inverter **14** into the through hole **64**.

(50) The motor wire **67** is drawn from the coil **58** of the motor **13**. That is, the motor wire **67** is drawn from the motor **13**. The cluster block **68** is made of resin. The cluster block **68** is formed in a rectangular box shape. A connecting terminal **68a** is housed in the cluster block **68**. The second end of the conductive member **66** is connected to the connecting terminal **68a**. The motor wire **67** drawn from the coil **58** is also connected to the connecting terminal **68a**. Thus, the conductive member **66** and the motor wire **67** are electrically connected to each other through the connecting terminal **68a**.

(51) The cluster block **68** is disposed in the through hole **64**. Accordingly, the wiring unit **65** is disposed in the housing **15** and inserted in the through hole **64**. Thus, the motor **13** and the inverter **14** are electrically connected to each other through the wiring unit **65**. Power regulated by the inverter **14** is supplied to the coil **58** through the conductive member **66**, the connecting terminal **68a**, and the motor wire **67** to rotate the rotor **56**. As a result, the rotary shaft **11** rotates integrally with the rotor **56**.

(52) <Flow of Refrigerant>

(53) In the centrifugal compressor **10**, when the rotary shaft **11** rotates, the refrigerant is sucked into the first impeller chamber **32** from the outside of the housing **15** through the suction passage **63**, the suction chamber **17**, and the first suction port **31** with rotation of the first impeller **16**. The refrigerant sucked into the first impeller chamber **32** is sent from the first impeller chamber **32** to the first diffuser **18** by centrifugal operation with the rotation of the first impeller **16**, and then, pressure of the refrigerant is increased in the first diffuser **18**. After that, the refrigerant having passed through the first diffuser **18** is discharged to the first volute **19**.

(54) The refrigerant discharged to the first volute **19** flows into the motor chamber S1 through the first communication hole **50**. The refrigerant in the motor chamber S1 is sucked into the second impeller chamber **38** through the second communication holes **51**, the bearing accommodating chamber **35**, and the second suction port **37** with rotation of the second impeller **41**. The refrigerant sucked into the second impeller chamber **38** is sent to the second diffuser **46** from the second

impeller chamber **38** by centrifugal operation with the rotation of the second impeller **41**, and then, pressure of the refrigerant is increased in the second diffuser **46**. After that, the refrigerant having passed through the second diffuser **46** is discharged to the second volute **47**.

(55) The refrigerant discharged to the second volute **47** is sucked into the third impeller chamber **40** through the third communication hole **52**, the communication passage **54**, and the third suction port **39** with rotation of the third impeller **42**. The refrigerant sucked into the third impeller chamber **40** is sent to the third diffuser **48** from the third impeller chamber **40** by centrifugal operation with the rotation of the third impeller **42**, and then, pressure of the refrigerant is increased in the third diffuser **48**. After that, the refrigerant having passed through the third diffuser **48** is discharged to the third volute **49**. The refrigerant discharged to the third volute **49** is discharged to the outside of the housing **15** and supplied to fuel cells.

Operation of the Embodiment

(56) The following will describe an operation of the present embodiment.

(57) Heat generated from the inverter **14** is dissipated to the refrigerant in the suction chamber **17** through the suction housing member **23**. As a result, the inverter **14** is cooled by the refrigerant in the suction chamber **17**. In addition, heat is dissipated from the refrigerant passing through the first diffuser **18** and the refrigerant discharged to the first volute **19** through the outlet of the first diffuser **18** with the rotation of the first impeller **16** to the refrigerant in the suction chamber **17**. Accordingly, each of the heat of the refrigerant passing through the first diffuser **18** and the heat of the refrigerant discharged to the first volute **19** through the outlet of the first diffuser **18**, with the rotation of the first impeller **16**, is hardly transmitted to the inverter **14** through the suction housing member **23**.

Advantageous Effect of the Embodiment

(58) The above-described embodiment provides the following advantageous effects.

(59) (1) The through hole **64** extends in the axial direction of the rotary shaft **11** to the inside of the inverter housing member **24** from the motor housing member **21** through the impeller housing member **22** and the suction housing member **23**. The wiring unit **65** is disposed in the housing **15** and inserted in the through hole **64**, and thus, the motor **13** and the inverter **14** are electrically connected to each other through the wiring unit **65** without extending the wiring unit **65** outside the housing **15**. Thus, unlike in a case where the wiring unit **65** extends outside the housing **15**, in the present embodiment, a seal member between the wiring unit **65** and the housing **15** is not required. Furthermore, unlike in the case where the wiring unit **65** extends outside the housing **15**, in the present embodiment, the wiring unit **65** does not need to be fixed to the housing **15** such that the wiring unit **65** does not interfere with auxiliary equipment, or the like disposed around the centrifugal compressor **10**. Therefore, in the present embodiment, the number of components of the centrifugal compressor **10** is not increased and ease of installation of the centrifugal compressor **10** is not reduced.

(60) In addition, the through hole **64** is disposed away from the first volute **19**, and the suction passage **63** is located away from the through hole **64** in the circumferential direction of the rotary shaft **11**. With this configuration, even when the through hole **64** through which the wiring unit **65** is inserted is formed so as to extend through the impeller housing member **22** and the suction housing member **23** in the axial direction of the rotary shaft **11**, the through hole **64** does not affect design of each of the first volute **19** and the suction passage **63** that are the existing configuration. Accordingly, in order to form the through hole **64** so as to extend through the impeller housing member **22** and the suction housing member **23** in the axial direction of the rotary shaft **11**, for example, positions of the first volute **19** and the suction passage **63** do not need to be changed. Thus, it is prevented that the centrifugal compressor **10** is increased in size by changing the positions of the first volute **19** and the suction passage **63**. Therefore, the motor **13** and the inverter **14** are electrically connected to each other without increasing the number of components and reducing the ease of installation of the centrifugal compressor **10** while the centrifugal compressor

10 is prevented from increasing in size.

(61) (2) The through hole **64** is disposed on the extension line of the discharge passage **61** extending in the circumferential direction of the rotary shaft **11**. This prevents the centrifugal compressor **10** from increasing in size as compared with a case where, for example, the through hole **64** is disposed outward with respect to the extension line of the discharge passage **61** extending in the circumferential direction of the rotary shaft **11** in the radial direction of the rotary shaft **11**. Accordingly, the motor **13** and the inverter **14** are electrically connected to each other without increasing the number of components and reducing the ease of installation of the centrifugal compressor **10** while the centrifugal compressor **10** is prevented from increasing in size.

(62) (3) When viewed in the axial direction of the rotary shaft **11**, a portion of the impeller housing member **22** that is located on the same side as the discharge passage **61** with respect to the imaginary straight line **L10** and adjacent to the downstream end of the discharge passage **61** in the circumferential direction of the rotary shaft **11** is a dead space. Here, when viewed in the axial direction of the rotary shaft **11**, the through hole **64** is located on the same side as the discharge passage **61** with respect to the imaginary straight line **L10** and adjacent to the downstream end of the discharge passage **61** in the circumferential direction of the rotary shaft **11**. Accordingly, the dead space of the impeller housing member **22** is effectively used as a space where the through hole **64** is formed. Thus, there is no need to separately provide a space where the through hole **64** is formed in the impeller housing member **22**, and thus, it is prevented that the centrifugal compressor **10** is increased in size by forming the through hole **64**.

(63) (4) When viewed in the axial direction of the rotary shaft **11**, the suction chamber **17** is overlapped with the first impeller **16**, the first diffuser **18**, and the first volute **19** such that the suction chamber **17** covers them. With this configuration, heat is easily dissipated from the refrigerant passing through the first diffuser **18** and the refrigerant discharged to the first volute **19** through the outlet of the first diffuser **18** with the rotation of the first impeller **16** to the refrigerant in the suction chamber **17**. Accordingly, each of the heat of the refrigerant passing through the first diffuser **18** and the heat of the refrigerant discharged to the first volute **19** through the outlet of the first diffuser **18**, with the rotation of the first impeller **16**, is hardly transmitted to the inverter **14** through the housing **15**. As a result, the reliability of the inverter **14** is improved. Furthermore, volume of the suction chamber **17** is made as large as possible, and thus, pulsations of the refrigerant sucked into the suction chamber **17** are reduced.

(64) (5) The cluster block **68** is disposed in the through hole **64**. With this configuration, the through hole **64** is effectively used as a space where the cluster block **68** being the existing configuration is disposed. Accordingly, there is no need to separately provide a space where the cluster block **68** is disposed, so that the size of the centrifugal compressor **10** is reduced.

Modification

(65) The above-described embodiment may be modified as follows. The above-described embodiment and the following modifications may be combined with each other as long as they do not technically contradict each other.

(66) In the embodiment, the through hole **64** may be located, for example, outward with respect to the extension line of the discharge passage **61** extending in the circumferential direction of the rotary shaft **11** in the radial direction of the rotary shaft **11**.

(67) In the embodiment, when viewed in the axial direction of the rotary shaft **11**, the through hole **64** may be located on the opposite side to the discharge passage **61** across the imaginary straight line **L10**.

(68) In the embodiment, when viewed in the axial direction of the rotary shaft **11**, for example, the suction chamber **17** does not need to cover a part of the first volute **19** and to be overlapped with the part of the first volute **19**.

(69) In the embodiment, when viewed in the axial direction of the rotary shaft **11**, for example, the suction chamber **17** does not need to cover a part of the first diffuser **18** and to be overlapped with

the part of the first diffuser **18**.

(70) In the embodiment, the cluster block **68** does not need to be disposed in the through hole **64**, and may be disposed, for example, in the motor chamber **S1**. Even in this case, the conductive member **66** is inserted in the through hole **64**. After all, the wiring unit **65** only needs to be inserted in the through hole **64**.

(71) In the embodiment, the centrifugal compressor **10** may have a configuration not including the third impeller **42**.

(72) In the embodiment, the centrifugal compressor **10** may have a configuration not including the second impeller **41** and the third impeller **42**.

(73) In the embodiment, the fluid to be compressed with the rotation of each of the first impeller **16**, the second impeller **41**, and the third impeller **42** is not limited to the refrigerant. Accordingly, any object to which the centrifugal compressor **10** is applied and any fluid to be compressed by the centrifugal compressor **10** may be used. For example, the centrifugal compressor **10** may be mounted on a fuel cell vehicle. In the fuel cell vehicle, a fuel cell system that supplies oxygen and hydrogen to a fuel cell to generate electricity is mounted on the fuel cell vehicle. Then, the centrifugal compressor compresses air as the fluid that includes oxygen to be supplied to the fuel cell vehicle. Furthermore, for example, the centrifugal compressor **10** may be applied to a refrigerant circuit for adjusting a temperature of a battery. In addition, an object on which the centrifugal compressor **10** is mounted is not limited to the vehicle and any object may be used.

SUPPLEMENTARY NOTES

(74) The following will describe technical ideas obtained from the above-described embodiments and the modifications.

Supplementary Note 1

(75) A centrifugal compressor comprising: a rotary shaft, a compression mechanism that compresses fluid with rotation of the rotary shaft; a motor that rotates the rotary shaft; an inverter that drives the motor; a housing in which the motor, the compression mechanism, and the inverter arranged in this order in an axial direction of the rotary shaft are accommodated; and a wiring unit through which the motor and the inverter are electrically connected to each other, the compression mechanism including: an impeller that rotates integrally with the rotary shaft; a suction chamber that is located closer to the inverter than the impeller in the axial direction; a diffuser that is disposed downstream of the impeller in a direction in which the fluid flows; and a volute that communicates with an outlet of the diffuser and is located outward with respect to the diffuser in a radial direction of the rotary shaft, characterized in that the housing has: a motor housing member in which the motor is accommodated; an impeller housing member in which the impeller is accommodated and that defines the diffuser and the volute; a suction housing member that defines the suction chamber; and an inverter housing member in which the inverter is accommodated, the motor housing member, the impeller housing member, the suction housing member, and the inverter housing member are arranged in this order in the axial direction, the housing has a through hole that extends in the axial direction to an inside of the inverter housing member from the motor housing member through the impeller housing member and the suction housing member, the wiring unit is disposed in the housing and inserted in the through hole, the through hole is disposed away from the volute, and a suction passage through which the suction chamber is connected to an outside of the housing is located away from the through hole in a circumferential direction of the rotary shaft.

Supplementary Note 2

(76) The centrifugal compressor according to supplementary note 1, characterized in that the volute has: a scroll passage that extends spirally around an axial line of the rotary shaft in the circumferential direction; and a discharge passage that branches off from the scroll passage, the discharge passage being located outward with respect to the scroll passage in the radial direction of the rotary shaft and extending around the axial line of the rotary shaft in the circumferential

direction, and the through hole is disposed on an extension line of the discharge passage extending in the circumferential direction.

Supplementary Note 3

(77) The centrifugal compressor according to supplementary note 2, characterized in that when viewed in the axial direction, a straight line intersecting the axial line of the rotary shaft and passing through a branch point at which the discharge passage branches off from the scroll passage is defined as an imaginary straight line, and when viewed in the axial direction, the through hole is located on the same side as the discharge passage with respect to the imaginary straight line and adjacent to a downstream end of the discharge passage in the circumferential direction.

Supplementary Note 4

(78) The centrifugal compressor according to any one of supplementary notes 1 to 3, characterized in that when viewed in the axial direction, the suction chamber is overlapped with the impeller, the diffuser, and the volute such that the suction chamber covers the impeller, the diffuser, and the volute.

Supplementary Note 5

(79) The centrifugal compressor according to any one of supplementary notes 1 to 4, characterized in that the wiring unit has: a conductive member that extends from the inverter into the through hole; a motor wire that is drawn from the motor; and a cluster block in which a connecting terminal through which the conductive member and the motor wire are electrically connected is housed, and the cluster block is disposed in the through hole.

Claims

1. A centrifugal compressor comprising: a rotary shaft; a compression mechanism that compresses fluid with rotation of the rotary shaft; a motor that rotates the rotary shaft; an inverter that drives the motor; a housing in which the motor, the compression mechanism, and the inverter arranged in this order in an axial direction of the rotary shaft are accommodated; and a wiring unit through which the motor and the inverter are electrically connected to each other, the compression mechanism including: an impeller that rotates integrally with the rotary shaft; a suction chamber that is located closer to the inverter than the impeller in the axial direction; a diffuser that is disposed downstream of the impeller in a direction in which the fluid flows; and a volute that communicates with an outlet of the diffuser and is located outward with respect to the diffuser in a radial direction of the rotary shaft, wherein the housing has: a motor housing member in which the motor is accommodated; an impeller housing member in which the impeller is accommodated and that defines the diffuser and the volute; a suction housing member that defines the suction chamber; and an inverter housing member in which the inverter is accommodated, the motor housing member, the impeller housing member, the suction housing member, and the inverter housing member are arranged in this order in the axial direction, the housing has a through hole that extends in the axial direction to an inside of the inverter housing member from the motor housing member through the impeller housing member and the suction housing member, the wiring unit is disposed in the housing and inserted in the through hole, the through hole is disposed away from the volute, and a suction passage through which the suction chamber is connected to an outside of the housing is located away from the through hole in a circumferential direction of the rotary shaft.

2. The centrifugal compressor according to claim 1, wherein the volute has: a scroll passage that extends spirally around an axial line of the rotary shaft in the circumferential direction; and a discharge passage that branches off from the scroll passage, the discharge passage being located outward with respect to the scroll passage in the radial direction of the rotary shaft and extending around the axial line of the rotary shaft in the circumferential direction, and the through hole is disposed on an extension line of the discharge passage extending in the circumferential direction.

3. The centrifugal compressor according to claim 2, wherein when viewed in the axial direction, a

straight line intersecting the axial line of the rotary shaft and passing through a branch point at which the discharge passage branches off from the scroll passage is defined as an imaginary straight line, and when viewed in the axial direction, the through hole is located on the same side as the discharge passage with respect to the imaginary straight line and adjacent to a downstream end of the discharge passage in the circumferential direction.

4. The centrifugal compressor according to claim 1, wherein when viewed in the axial direction, the suction chamber is overlapped with the impeller, the diffuser, and the volute such that the suction chamber covers the impeller, the diffuser, and the volute.

5. The centrifugal compressor according to claim 1, wherein the wiring unit has: a conductive member that extends from the inverter into the through hole; a motor wire that is drawn from the motor; and a cluster block in which a connecting terminal through which the conductive member and the motor wire are electrically connected is housed, and the cluster block is disposed in the through hole.
